



Context Driven Multi-query Resolution Using LLM-RAG to Support the Revision of Explainability Needs

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Abstract. The revision step in the Case-Based Reasoning (CBR) cycle ensures that cases are adaptable and that updates can be integrated meaningfully based on evaluation metrics. However, the effectiveness of this step heavily depends on how new knowledge is acquired to support revision. In the iSee project, where CBR is used for explanation strategy recommendations, revision knowledge is typically derived from end-user feedback following an interactive, user-centric explanation experience. This raises the research question: how can we discover this knowledge from user interactions, and how can Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) enhance the explanation experience while capturing useful knowledge for the discovery? In this paper, we propose a methodology for detecting the evolution of user explanation intent (an indicator for the need for revision) through LLM-RAG enhanced interactions. Finally, experimental results evaluate the success of the proposed methodology, assessing its validity and, in particular, the reasonability of the LLM-RAG approach. Experimental results across multiple real-world use cases demonstrate that our methodology produces highly coherent and context-aware explanations, improving overall explanation clarity, and effectively identifies when explanation strategies require revision.

Keywords: Interactive Explanations · Large Language Model · Reasonability · Explainability Intent · Explanation Experience

1 Introduction

Explainable Artificial Intelligence (XAI) plays a key role in ensuring that we can place trust in AI decisions when they impact crucial areas, such as predicting the presence of a tumour [2], making a decision in a trial [21], or detecting possible attacks on citizen security [28]. Since early 2020, legal regulations framework have mandated that AI-based systems must produce meaningful, transparent

and accurate explanation for the results they expose to human users [16]. However such explanations are generally multifaceted, as users' intents span across multiple explanation needs including task performance improvement, actionability, trust, learning, or transparency. Evaluating the quality of these explanations remains a challenge [8]. To achieve this, the iSee platform [3] incorporates methods for assessing the effectiveness of an explanation strategy tailored to AI scenarios, alongside knowledge structures that describe the user context. These knowledge structures enable systematic comparison of user context descriptions facilitating the identification of similar scenarios, the understanding of differences in context, and exploiting it to manage adaptation to variations in user explanation needs. The explanation strategies are collected through the development of several real-world use cases. In this way, the generated catalog of strategies can be transferred to other AI solutions. From this catalog, iSee applies a Case-Based Reasoning (CBR) [1] methodology to retrieve the most suitable explanation strategy that it has learned for the given domain and task and then adapts it to meet the end-user's requirements.

Work combining CBR and LLM have started to emerge over recent years [26]. LLMs, with their ability to process and generate human-like text, can enrich CBR systems by providing more natural language explanations, however ensuring that the LLM generates grounded, complete and coherent response remains a challenge. For this reason LLM based Retrieval-Augmented Generation (RAG) is used to refine the contextual knowledge used when generating responses. To assess the coherence, relevance and clarity of the response provided by the LLM-RAG, the Reasonability measure has gained attention as a useful means to ensure that the integration of external knowledge sources to LLM-RAG does not compromise the validity of the response and helps to evaluate how the responses align with human judgment [11].

Previous research work studied the benefits of iSee in evaluating the quality of explanations [3]. The findings indicate that using a structured, hierarchical structure, such as a behavior tree is effective in managing interactive explanation dialogues. They also highlight the need for more flexible conversation turns to accommodate user clarification needs. Accordingly, this paper makes the following contributions to iSee and XAI-CBR literature.

- An enhanced explanation strategy model with open free-text questions to allow users to express their clarification needs in their own words.
- An LLM-RAG approach that guides clarification responses interactive XAI by providing relevant context tailored to the end-user's experience. Specifically the context uses the iSee ontology and the instantiated knowledge accumulated through the traversal of the behaviour tree during the Explanation Experience.
- A metric to evaluate the reasonableness of LLM-RAG generated responses.
- A revision knowledge discovery method that identifies evolving user intent from end-user interactions to inform the revision of explanation strategies.

The rest of the paper presents the interactive clarification method in Sect. 3, and a methodology to assess the relevance of the new knowledge discovered from

iSee clarification interactions with end users. Experimental results on three real-world use cases and one academic use case are described in Sect. 4 and conclusion and future work are discussed in Sect. 5.¹

2 Related Work

The iSee platform enables AI developers to reuse satisfactory explanation strategies for a given AI scenario [12, 23, 25]. Design-users define the contextual requirements, including the domain, dataset, model to be explained, as well as the different stakeholder groups), and End Users interact through a conversational dialog manager (chatbot) with the different explainer algorithms within the strategy, and provide evaluation feedback about their explanation experience. This user-centered feedback helps Design Users refine the original explanation strategy to enhance the explanation experience. All aspects of the experience (the contextual requirements supplied by the design-user, the recommended explanation strategy, and end-user feedback) are then stored as a case to support reuse and revision tasks of the CBR iSee cycle [3].

An explanation strategy in iSee framework, i.e., the case solution is modelled using a Behaviour Tree (BT) [4]. An example of explanation strategy using Behaviour Tree is detailed in [3]. This tree structure is a Directed Acyclic Graph (DAG), formed of multiple user question-explanation strategy sub-trees, to enable the reuse of common sub-trees without duplicating nodes. Each sub-tree is formed on the basis of an “intent” question category and a combination of explanation methods. In this work we introduce a new concept into the Behavior Tree to represent the need for clarifications expressed by the end user. For example, within a behaviour tree, if a factual explainer such as LIME addresses a user’s initial intent question (“Why did I receive this outcome?”), a linked clarification node can handle subsequent queries, such as “How were those attribution weights computed?”. At which point we can trigger a LLM-RAG response using an LLM. In this way, a BT provides a structured and controlled approach, enabling systematic handling of various types of user questions by mapping them either to explainer nodes or LLM-RAG clarification nodes.

It is particularly important to prove that introducing LLM-RAG does not compromise the quality of explanation strategies, while ensuring that the clarifications provided about iSee explanations knowledge are complete and aligned with user requests. There is a rising trend toward using LLMs as judges for evaluation due to alignment performed on LLMs on human feedback [15, 29] and human evaluations are often not scalable enough for complex, high-volume tasks. Recent studies have demonstrated that LLM-based evaluators can provide consistent, context-aware assessments while bypassing the high costs and limited throughput of manual human judgment. These approaches leverage in-context learning and advanced prompt engineering so that the LLMs can mimic human-like reasoning when scoring outputs [11]. In our work we will study how to apply

¹ Results and code used in the paper are available at <https://github.com/LasalJayawardena/Context-Driven-Multi-Query-Resolution/>.

this task to the domain of explanation experience sharing by human users and for human users, and to which level the metrics facilitate the evaluation of clarity and coherence of LLM-generated responses in iSee.

3 Methodology

In this section, we formalise the proposed methodology that enhances the *iSee* platform with large LLM capabilities for interactive clarifications and adaptive explanation strategies. The approach draws upon established principles from the CBR cycle, especially focusing on the *revision* of explanatory knowledge, and augments a previously static Behaviour Tree with a new Clarification Node driven by an agentic workflow. We also introduce the notion of a *Reasonability Score* to quantitatively assess explanation quality, and employ a Markov chain analysis to capture and model user intent evolution.

3.1 Overview of the iSee CBR Cycle

The *iSee* platform follows a CBR-inspired cycle for designing and refining explanation strategies. Formally, let

$$\mathcal{E} = \{XS_1, XS_2, \dots, XS_m\} \quad (1)$$

be the set of available *explanation strategies* (XS) in the system. Each strategy XS_k is an instantiated flow of explanatory components tailored to particular user personas and goals. The CBR cycle typically involves:

1. **Retrieve:** Selecting the most relevant $XS_k \in \mathcal{E}$ based on the user’s persona and intent.
2. **Reuse:** Adapting XS_k to address specific user queries.
3. **Revise:** Incorporating new insights (e.g., user feedback or clarifications) to refine XS_k .
4. **Retain:** Storing improved XS_k in the repository of explanation strategies for future reuse.

While previous versions of *iSee* relied on a *XEQ* (“seek”) *Questionnaire* [24] and manual inspection to identify when to revise XS_k , this work introduces the *Clarification node* in the behaviour tree, enabling an interactive clarification process that enables the implicit discovery of revision knowledge applicable to the explanation strategy. Whenever user intent changes, the Clarification node drives the CBR **Revise** step: each clarification exchange is captured as revision knowledge and can be integrated back into the explanation strategy, updating the behaviour-tree’s interaction to better match the user’s evolving needs in the future.

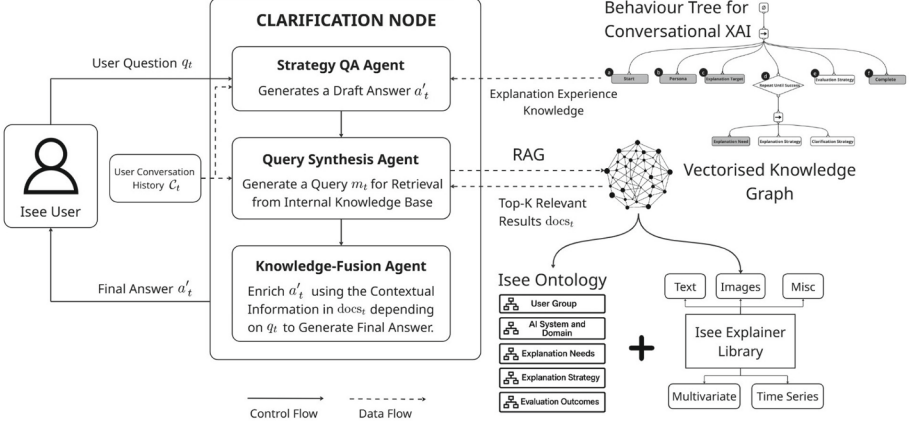


Fig. 1. An illustration of the iSee LLM Clarification Node agentic flow.

3.2 Clarification Node in the Behaviour Tree

A *Behaviour Tree* T defines the conversational and explanatory flow of *iSee*. In prior work, T was relatively static, offering limited user-driven revisions. We denote T as a directed acyclic graph of nodes:

$$T = (N, E), \quad (2)$$

where $N = \{n_1, n_2, \dots, n_\ell\}$ is the set of nodes and E is the set of directed edges encoding transitions. We introduce a special node $n_{\text{clar}} \in N$, which is triggered whenever the user requests additional detail or signals uncertainty. Formally, we define a trigger function Φ that transitions from a standard explanatory node n_j to n_{clar} :

$$\Phi : N \times \{\text{user feedback}\} \mapsto n_{\text{clar}} \quad (\text{if clarification needed}). \quad (3)$$

Once at n_{clar} , the system spawns an *agentic workflow* (see Fig. 1). The user may engage in multiple rounds of clarification until satisfied, denoted by $q_0, q_1, \dots, q_r$, where q_t is the t -th clarification question. The agentic workflow draws on state of art agentic design methodologies [22] and utilises *three agents*, underpinned by a shared instruct-tuned LLM, as follows:

1. **Strategy QA Agent:** This agent leverages the Behaviour Tree T (and any conversation state C_t) to generate an initial, context-aware *draft* answer:

$$a'_t = \text{StrategyQA}(q_t, T, C_t). \quad (4)$$

Here, a'_t aligns with the user's intent as captured in T . It does not incorporate any newly retrieved knowledge, relying solely on the conversation history and previous interactions in T .

2. **Query Synthesis Agent:** The system next identifies potential additional knowledge from the iSee knowledge bases $\mathcal{K}$, which is a collection of internal iSee usecase, explainers and dataset information, that may enrich or clarify the draft answer. The Query Synthesis Agent extracts metadata such as dataset types, relevant explainer methods, or domain facts from the user query q_t and relevant context C_t into a structured format for retrieval. The metadata may include the *dataset categories* (e.g., “image,” “tabular”), recommended *explainers* (e.g., SHAP, LIME, GradCAM), or domain-specific facts (e.g., “X-ray guidelines”). This process is guided by the few-shot examples E_{fewshot} which is a representative collection of examples of potential metadata that could be extracted:

$$m_t = \text{QuerySynthesis}(q_t, C_t, E_{\text{fewshot}}). \quad (5)$$

The system uses m_t along with the original query q_t to call a two-step RAG function:

$$\text{docs}_t = \text{RAG}(m_t, q_t, \mathcal{K}), \quad (6)$$

which retrieves the top- k relevant results from $\mathcal{K}$. The system applies an empirically selected semantic similarity threshold, based on cosine similarity with the user query q_t , to filter out less relevant items. In addition, the metadata m_t filters or reorders candidate results, further improving retrieval accuracy. If no relevant external context is found, docs_t remains empty.

3. **Knowledge-Fusion Agent:** Finally, the system optionally fuses any newly retrieved information (docs_t) with the draft answer a'_t :

$$a_t = \text{KnowledgeFusion}(a'_t, \text{docs}_t, q_t), \quad (7)$$

Even when docs_t is non-empty, the $\text{KnowledgeFusion}(\cdot)$ function may still opt to discard or ignore certain information. Specifically, it inspects each item in docs_t to determine whether it adds clarity or usefulness to the user’s query q_t . If the retrieved content proves relevant, it is merged into a'_t ; otherwise, the agent simply returns a'_t unchanged, ensuring that output formatting and other alignment constraints are satisfied.

Hence, for each clarification query q_t , the final response a_t follows the chain:

$$q_t \xrightarrow{\text{Strategy QA}} a'_t \xrightarrow{\text{Query Synthesis}} m_t \xrightarrow{\text{RAG}} \text{docs}_t \xrightarrow{\text{Knowledge-Fusion}} a_t$$

This workflow ensures that the user’s clarification remains centered on the original explanation path while only selectively incorporating external knowledge. The motivation for using a multi-agent workflow is that relying on a single agent for all tasks can cause the agent to deviate from paying attention to relevant parts of the Behavior Tree T and conversation context C_t , often producing more generic or less context-specific responses. Moreover, repeated or unexpected reliance on the *Query Synthesis Agent* can signal that the user’s *explanation intent* has diverged from the originally designed strategy, prompting the need to track these changes. Such shifts are formally detected to trigger the *revision* step of the CBR cycle, where the explanation strategy is updated to reflect newly discovered user needs.

3.3 Revision Knowledge Discovery

To quantify the clarity and utility of each generated clarification, we propose a *Reasonability Score* R . Reasonability Score is defined in the Literature [11] as a measure comparing clarity of LLM responses. The idea is to use other large language models as impartial judges. Suppose an answer a_t is produced in response to the clarification query q_t under conversation context $\mathcal{C}_t$ and with access to knowledge base $\mathcal{K}$. We define:

$$R(a_t) = \frac{1}{|M|} \sum_{m \in M} \text{Score}_m(q_t, \mathcal{C}_t, \mathcal{K}, a_t), \quad (8)$$

where M is a set of LLM *judges*, and $\text{Score}_m(q_t, \mathcal{C}_t, \mathcal{K}, a_t) \in [0, 10]$ is the individual judge’s rating of how coherent, correct, and context-aligned the answer a_t is, given the query q_t , the conversation context $\mathcal{C}_t$, and relevant knowledge base information $\mathcal{K}$. We typically use $|M| \geq 30$ for robust averaging. In practice:

1. Each model $m \in M$ is prompted with q_t , $\mathcal{C}_t$, $\mathcal{K}$, and a_t , and asked to produce a numerical rating reflecting the answer’s coherence, factual correctness, and alignment with the conversation context and knowledge base.
2. The average $R(a_t)$ over all models in M is taken as the final *Reasonability Score*.
3. A 95% confidence interval is computed to measure variability across different judging models.

This score serves as a core part of the *revision knowledge* in the CBR cycle, informing whether XS_k should be updated to produce clearer or more comprehensive explanations. The application of Reasonability score methodology to iSee platform will be described under the Evaluation section.

Beyond individual clarifications, users’ *intents* often shift during the conversation. We denote the set of possible user intents as

$$\mathcal{I} = \{\iota_1, \iota_2, \dots, \iota_n\}. \quad (9)$$

At each clarification step t , we observe a random variable X_t that represents the user’s intent:

$$X_t \in \mathcal{I}, \quad (10)$$

with $X_t = \iota_i$ meaning the user’s intent at step t is ι_i . As clarifications progress ($t = 0, 1, \dots, r$), we track X_t either through explicit user statements or inferred signals (e.g. from the user’s query content).

Transition Dynamics. We model the intent evolution $\{X_t\}$ as a discrete-time Markov chain with transition probabilities

$$P_{ij} = \Pr(X_{t+1} = \iota_j \mid X_t = \iota_i), \quad (11)$$

where P_{ij} is the probability of moving from intent ι_i to ι_j in one clarification step.

Over multiple user sessions, we estimate P_{ij} empirically by counting intent transitions:

$$\hat{P}_{ij} \approx \frac{\text{Number of transitions } (\iota_i \rightarrow \iota_j)}{\sum_{k=1}^n \text{Number of transitions } (\iota_i \rightarrow \iota_k)}. \quad (12)$$

In many cases, certain states (e.g. *transparency*) may be more recurrent or lead to frequent transitions (e.g. *transparency* $\rightarrow$ *debugging*), reflecting the evolving needs of the user.

Finally, the platform unifies:

1. **Clarification Node Interactions:** $\{q_0, q_1, \dots, q_r\}$ which is the set of clarification queries and their corresponding answers $\{a_0, a_1, \dots, a_r\}$, each evaluated with a *Reasonability Score* $R(a_t)$.
2. **User Intent Evolution:** The Markov chain transitions P_{ij} derived from user sessions.

Collectively, these form the *revision knowledge* needed to update or replace XS_k . Formally, let

$$\Omega(\text{XS}_k, \{q_t, a_t, R(a_t)\}_{t=0}^r, P) \mapsto \text{XS}'_k, \quad (13)$$

be the *revision function*, which outputs an updated strategy XS'_k by:

1. *Incorporating* high-level user requests identified in q_t .
2. *Adjusting* explanatory steps that yield persistently low $R(a_t)$ scores.
3. *Realigning* to user intents suggested by the transition probabilities P_{ij} .

This end-to-end workflow can automate the adaptation of explanation strategies to evolving user needs, while preserving the core CBR principle of *learning from experience*.

4 Evaluation and Use Cases

Previous experiments revealed that the explanation strategy used by the iSee chat interface was not always presented in a manner that was easily understood by humans. This highlighted the need for more readable explanations and the ability for end-users to request additional details about the explanation or the result.

We evaluated the effectiveness of the tools presented in this paper by elaborating relevant elements that align with user clarification requests regarding explanation strategies, and in identifying when user requirements or intents evolve during conversations.

The research questions are:

- **Meaningfulness:** Do users perceive the explanation conversational experience as more qualitative when using the clarification step?
- **Adaptation of explanation needs:** Can we construct a task that detects changes in explanation strategy goal and allow revising the explanation strategy context based on end user feedback?

We conducted evaluation experiments with a diverse group of 10 participants, including engineers, managers, and desk agents. Although not all were experts in AI or the specific domain used in the experiment, they all had experience in information technology and a technical background. The participants were asked to use iSee tools in one to four real-world use cases.

To address ethical concerns, we informed the users about the nature of the experiment, its goals, the tasks they needed to perform, and how the results would be used. No personal data was collected, ensuring the participants' autonomy, privacy, and confidentiality, and providing them with trustworthy information. The study followed the same protocol as described in [3].

4.1 Summary of the Use Cases

3 real world and 1 academic use cases have been proposed to the end user community of 10 people. The use cases are from different industrial sectors:

- **Telecommunication sector:** The AI task is the classification of field work scenario based on engineering task notes. The goal of the AI system is to recommend the most suitable next task to network planner from unstructured textual notes. 2 persona were designed: Telecom Planner where we set the goal of explanation to be understanding the reasoning behind the recommender (transparency) and Engineer where we set the goal of explanation to be evaluating whether the AI system works as expected (debugging).
- **Medical sector:** The AI task is the analysis of x-ray images from medical imaging equipment to support a medical expert or a manager. The goal of the AI system is to detect whether a fracture is present. 2 persona were designed: Clinician where we set the goal of the explanation to be supporting in assessing the result and deciding the next task (accountability) and Manager where we set the goal of the explanation to be understanding the reason why the AI system produces its result and how accurate the system is (transparency and performance).
- **Manufacturing sector:** The AI task is the analysis of equipment images to detect faulty produced assets in a production line. The goal of the AI model is to detect defects in the manufactured components using computer vision techniques. 2 Persona were designed: Auditor where we set the goal of explanation to be justifying the decision of acceptance or rejection of the component (actionability and transparency) and Engineer where the goal of explanation was set as informing about the goodness of the AI system (performance).
- **Finance Loan sector:** the AI task is the recommendation about the likelihood of a loan request to be approved based on requestor profile. One user was designed, with a goal of explanation set to improve trust in the AI system result.

4.2 Reasonability Evaluation Using LLMs

Given the limitations of conventional readability metrics, we adopted a more nuanced evaluation strategy. Our approach involved evaluating explanations using multiple LLM-generated Reasonability Scores, allowing nuanced judgment on the quality and relevance of explanations.

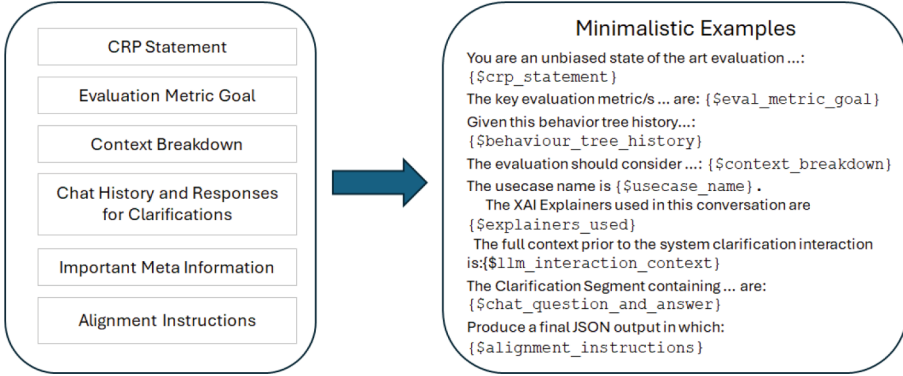


Fig. 2. Reasonability Assessment Prompt.

Evaluation Approach

- **Test Models:** In the evaluation, over 100 LLMs were tested, however, only 70 LLMs were comprehensively reported, as they were able to generate reasonability scores for all question-response pairs, whereas the others could not. From the total LLM collection 40% were closed source and 60% were open source. Open source LLM model variants such as LLAMA V1 [20], LLAMA V2 [20], LLAMA 3 [7], DeepSeek R1 [6], DeepSeek V2 [5], Flacon 3 Family², Gemma V1 [19], Gemma 2 [18], IBM Granite [13], Mistral [9], Mixtral [10], Nemotron [17], and Qwen 2.5 [27] were used. Closed source model variants such as OpenAI GPT-4 [14], OpenAI GPT-O reasoning models, Anthropic Claude³, Google Gemini (including reasoning variants)⁴, and Mistral⁵ were used.
- **Reasonability Score:** Each explanation was assessed using an LLM-as-a-Judge approach, with LLMs generating a quantitative reasonability score (0–10 scale) for each model. The prompt used for the evaluation is outlined in Fig. 2.

² <https://huggingface.co/blog/falcon3>.

³ <https://www.anthropic.com/claude>.

⁴ <https://gemini.google.com>.

⁵ <https://mistral.ai>.

- **Confidence and Impartiality Analysis:** To robustly determine fairness and consistency, median scores were calculated and impartial models were identified. Five randomly sampled questions and the relevant results are shown in Table 1. The most impartial LLM for all questions based on median was identified to be *Qwen-2.5-32B*.

Table 1. Reasonability Scores for 5 Randomly Sampled Questions asked

Question (Verbatim)	Avg with CI	Median	Std Dev
Don't give me the value in terms of the model parameters/features, give me the data value.	6.89 ± 0.47	7.00	2.01
Give me insights about how LIME explain that the result will be the same for similar instance	8.45 ± 0.15	8.00	0.63
Que signifie une influence negative sur le resultat? (<i>French in text</i>)	8.71 ± 0.20	9.00	0.84
What is the mat behind it	6.94 ± 0.46	8.00	1.97
Give me 2 other types of explanation of the result	8.71 ± 0.21	9.00	0.90

4.3 User Intent Evolution with Markov Chain Analysis

Beyond individual explanation clarity, our study also aimed to capture how user intents evolve over multiple clarification interactions. We collected user-chat sequences and labeled each clarification turn with a specific intent - the value being taken among the set of 16 possible intent values defined in iSee platform [8]. These labeled sequences were then used to construct a Markov chain model, where each state corresponds to an explanation intent, and transitions reflect how a user's intent changes from one clarification turn to the next.

Ordered Intent Sequences. Below is a sample of the how user intent shifts within a conversation:

- Chat 1: Transparency → Performance : Transparency
- Chat 2: Actionability : Education
- Chat 3: Transparency → Transparency : Transparency
- ⋮
- Chat 27: Performance → Transparency : Comprehensibility → Comprehensibility → Transparency
- Chat 28: Debugging : Comprehensibility
- Chat 29: Transparency : Transparency

The “:” indicates the starting point of the LLM clarification interaction.

In the examples above, we observe that on interaction 27, the end user shifted expectations from the explanations and clarifications produced by the tool. The intent then moved towards comprehensibility which indicates a need for structured and consistently formatted responses, while finally user was again focusing on transparency goal once having read more structured clarifications from the chatbot. Moreover in interaction 2, we observe that use case originally was designed with an actionable explanation need, whereas, user clarification questions suggest that end user was moving explanation needs toward a need for learning new knowledge about the explanation methods themselves.

Full Sequence Analysis: By aggregating all chats, we derived a transition matrix capturing the frequency with which one intent leads to another. Figure 3 depicts the probability transition matrix for the most frequently observed intents.

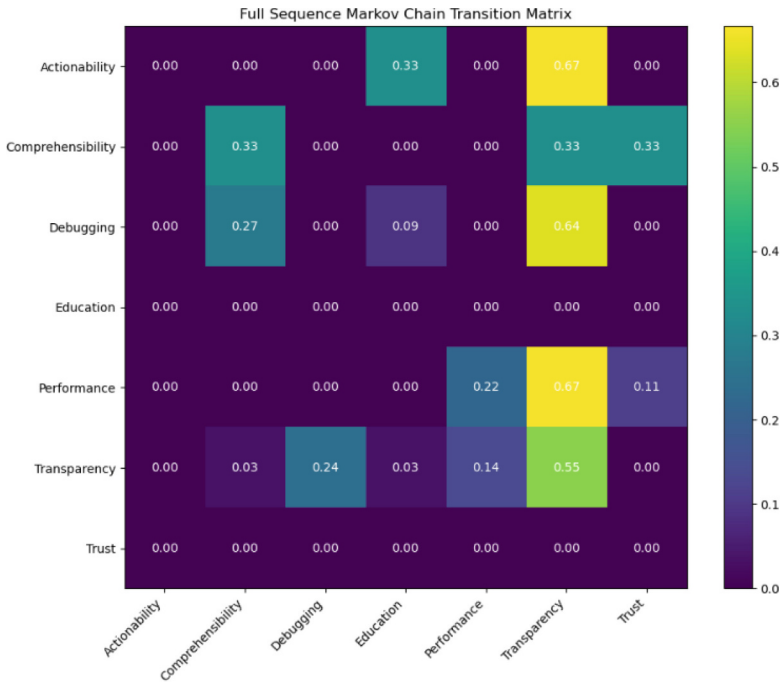


Fig. 3. Markov Chain Probability Transition Matrix.

4.4 Insights and Observations

A statistical analysis of the reasonability scores was performed, computing average scores, variance, standard deviation, and confidence intervals for each ques-

tion. In addition, Shannon Entropy, Simpson Index, and Gini Coefficient were also computed for sampling purposes. Notable insights include:

- **Average Scores** ranged from 3.31 (for a low-complexity question about sensor measurement) to as high as 9.05 (for a question regarding precision and recall). On average, the reasonability scores were closer to 8 out of 10 or more for the clarification questions asked by the user.
- The highest scoring explanations included detailed clarifications on concepts such as SHAP vs. LIME (average score: 9.03), suggesting users found comparative and well-structured explanations more reasonable.
- Explanations requiring basic conceptual understanding, such as “*what is fluidity*”, scored slightly lower (average 7.87), suggesting potential areas for improvement in simpler, foundational explanations.
- **Impartiality:** Models frequently identified as impartial included *Qwen-2.5-32b*, *Mistral-Large*, and *Anthropic-claude-3-sonnet*.

An additional Markov chain analysis of user intent transitions revealed that **Transparency** often serves as a central pivot: users repeatedly transition from and back to **Transparency**. Frequent moves between **Performance** and **Transparency** suggest that users initially seek general explanations, then deeper performance insights, and subsequently request clarifications about how decisions are made. The recurring transitions between **Debugging** and **Transparency** indicate that once users identify potential system issues, they often return to requesting transparent details for verification or deeper understanding.

These patterns show that clarifications lead users to explore multiple facets of explanation, ranging from performance-oriented queries to debugging or trust concerns, highlighting the importance of adaptive, multi-intent explanation strategies. One of the most interesting revelations is the evolution of original intents to intents which were not modelled by the design user, for example **Actionability** to **Education**.

5 Conclusions and Future Research Directions

We proposed a methodology that enhances the iSee platform with LLM-RAG capabilities for interactive clarifications and adaptive explanation strategies. We also introduced the Reasonability Score to quantitatively assess explanation quality and employ a Markov chain analysis to capture and model the evolution of user explanation needs (intent), ultimately informing the revision task of the CBR cycle of iSee. The resulting workflow sets the grounds for automating the adaptation of explanation strategies to evolving user needs, while preserving the core CBR principle of *learning from experience*.

Finally we analysed how the resulting LLM-RAG has helped clarifying the explanation produced by iSee and improved explanation experience of the end user overall. We analysed the groundness of the proposed methodology and especially the proposed LLM-RAG reasonability that we compared to 100 other

LLMs in the Literature. The evaluation experiments (4 use cases) underscores the need for sophisticated reasonability scoring using LLMs and suggest reasonability of our LLM-RAG is on the higher end. Markov chain analysis provided insights into user intent evolution, emphasising user demands for transparency and debugging-focused explanations. The overall methodology enabled the feasibility of discovering when end users change their intent.

This work opens more research directions related to the use of LLM with CBR systems, including 1) The auto-adaption of the explanation flow based on user interaction patterns, 2) the dynamic exploration of other explainers within the iSee dialog manager, and 3) Further investigation of open-source, smaller parameter models for scalable explainability systems.

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